Audit Trail — Case 11: Shoulder closure on divided highway

Field	Value
MUTCD typical application	TA-5
CDOT standard sheet	S-630-1
Case	Case 11: Shoulder closure on divided highway
Taper length	217 ft (L/3 (shoulder taper))
Buffer space	645 ft
Device spacing — taper	65 ft
Device spacing — tangent	130 ft
Crew steps	11
Routing	shoulder_no_reduction

Taper Length

Speed 65 mph >= 45 mph threshold -> using L = W x S

L = 10 x 65 = 650 ft

L/3 = 650 / 3 = 217 ft

Field	Value
Closure type	shoulder
Offset (shoulder width)	10 ft
Required taper (L/3 (shoulder taper))	217 ft

Source: MUTCD 11th Ed. Sec 6B.08, Table 6B-3. Shoulder closures use L/3 per Sec 6B.08 (Table 6B-3).

CDOT reference: CDOT S-630-1 Case 11 (right-shoulder closure on divided highway)

Buffer Space

MUTCD Table 6B-2: 645 ft. CDOT S-630-1 Sheet 14 posts a 570 ft minimum at 65 mph, but only as part of the Case 26 mandatory speed step-down (65 -> 60 mph); this plan has no qualifying reduction, so the federal posted-speed value governs (Sheet 7 Case 11: buffer VARIES per General Note 24).

Source: MUTCD 11th Ed. Sec 6B.06, Table 6B-2 (stopping sight distance). CDOT S-630-1 Sheet 7 Case 11 (buffer VARIES per General Note 24).

Channelizing Device Spacing

65 mph -> 65 ft spacing (MUTCD Sec 6K.01: spacing equals speed in feet)

65 mph -> 130 ft spacing (MUTCD Sec 6K.01: spacing equals 2x speed in feet)

L/3 = 217 ft / 65 ft max spacing -> 5 drums at 54.2 ft intervals

1000 ft / 130 ft max spacing -> 9 cones at 125.0 ft intervals

Downstream taper: 50 ft at ~25 ft spacing -> 2 cones (MUTCD Sec 6B.08: 50-100 ft downstream taper); 9 tangent + 2 downstream = 11 cones total

Source: MUTCD 11th Ed. Sec 6K.01

Advance Warning Signs

Speed 65 mph, road_type = 'freeway' -> Table 6B-1 category: freeway

A = 1000 ft, B = 1500 ft, C = 2640 ft

Position	Code	Station (ft)	Distance from Taper (ft)
C (furthest)	W20-1	7,002	5,140 upstream
B (middle)	W20-2	4,362	2,500 upstream
A (nearest)	W21-5aR	2,862	1,000 upstream
A plaque	W16-2a	2,862	NEXT 1,862 FT (under W21-5aR at A)
A2 (second W21-5aR)	W21-5aR	2,612	750 upstream
A2 plaque	W7-3a	2,612	NEXT 1 MILE (under second W21-5aR)
Supplemental (W5-1)	W5-1	2,362	500 upstream

Source: MUTCD 11th Ed. Table 6B-1

Colorado Requirements (CDOT S-630-1)

Check	Result	Citation	Detail
Signs on both sides of divided highway	PASS	CDOT S-630-1 (July 2026) Sheet 2, General Note 8	Required: True. Signs placed: 12 left, 12 right.
G20-5P Work Zone signs every 2,640 ft	PASS	CDOT S-630-1 (July 2026) Sheet 2, General Note 4	Zone length: 7,002 ft. Required: 3. Placed: 3.
Speed reduction <= 15 mph per sign installation	PASS	CDOT S-630-1 (July 2026) Sheet 2, General Note 3	No work-zone speed reduction. Posted speed 65 mph applies throughout the zone.
Flagger station lighting 500W @ 8 ft	PASS	CDOT S-630-1 (July 2026) Sheet 2, General Note 22	Not applicable (no flaggers).

AADT threshold for mobile operations (<= 2,000) — Not applicable (not a mobile operation). (CDOT S-630-1 (July 2026) Sheet 23, Case 38 Note 1)

All Colorado checks pass: True

CDOT Case Reference

Case 11: Shoulder closure on divided highway

This scenario matches CDOT Standard Plan S-630-1, Case 11: shoulder closure on a divided highway.

The generated plan follows the same device layout as the reference case with spacing computed for 65 mph.

Routing: shoulder_no_reduction

Reference:
https://www.codot.gov/safety/traffic-safety/assets/s-standard-plans/s-630-1/S-630-01%20(19-Page%20Set).pdf

Geometry Validation

Work zone 1000 ft vs required taper 217 ft (L/3 (shoulder taper)) + buffer 645 ft.

All geometry checks pass: True

Source: MUTCD 11th Ed. Sec 6B.06 (buffer) and Sec 6B.08 (taper)

Corridor Validation

OpenStreetMap corridor check ran with no warnings.

Flagger Control

Not applicable for this scenario.

Source: MUTCD 11th Ed. Chapter 6D (Flagger Control) and Chapter 6E (One-Lane, Two-Way Traffic Control).

Pending Verification

0 reference(s) pending verification. Tracking: https://github.com/rtmakatura/conestruct/issues/19